

Two routes to the variance conjecture for prime values of polynomials: the off-diagonal over $\mathbb{Z}$ and over $\mathbb{F}_q[u]$

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Abstract

Companion to *The variance of the number of prime values of a polynomial*. That paper reduces the conjecture $\sum_{h \leq H} (S_f(h) - C(f)^2) = -\frac{1}{2}C(f) \log H + O(1)$ to Hypothesis (E): the off-diagonal sum over pairs of distinct roots of f modulo d contributes no logarithm. Here we work on the two routes to proving (E). Over $\mathbb{Z}$, for quadratics, we show that (E) is exactly an average equidistribution statement for the roots of $x^2 \equiv D \pmod{d}$ of the type proved by Hooley, and we isolate the one estimate still needed (the large moduli, Section 2). Over $\mathbb{F}_q[u]$ we prove that the off-diagonal vanishes identically for all moduli of degree at most N , so that the analogue of the conjecture reduces to the moduli of degree $> N$ alone, where it becomes an equidistribution statement of Katz type; we verify the resulting exact identity numerically over $\mathbb{F}_3$, $\mathbb{F}_5$ and $\mathbb{F}_7$ (Section 3).

1 Notation and the identity to be proved

$f \in \mathbb{Z}[t]$ irreducible, without fixed prime divisor; $\omega_f(p)$ roots modulo p ; $\nu_f(d, h) = \#\{s \pmod{d} : d \mid f(s), d \mid f(s+h)\}$; $C = C(f)$; $S_f(h)$ the pair singular series; $b(d) = \prod_{p \mid d} (p - \omega_f(p))^{-2}$, $P_p = 1 - \omega_f(p)^2 / (p - \omega_f(p))^2$, $W_f(d) = db(d) \prod_{p \nmid d} P_p$, $a_f(d) = W_f(d) \omega_f(d)$ (d squarefree). The exact identity of the paper is

$$\sum_{h \leq H} (S_f(h) - C^2) = -C^2 \sum_d a_f(d) \left\{ \frac{H}{d} \right\} + C^2 \text{Off}_f(H), \quad \text{Off}_f(H) = \sum_d W_f(d) \sum_{s \neq s'} \psi_d(s' - s, H), \quad (1)$$

$\psi_d(m, H) = \#\{h \leq H : h \equiv m \pmod{d}\} - H/d$, the inner sum over ordered pairs of distinct roots of f modulo d . The diagonal is understood (Theorems 3–4 of the paper); **Hypothesis (E)** is $\text{Off}_f(H) = O(1)$, and its Cesàro form $\text{Off}_f^*(H) = O(H)$ gives the conjecture with the leading term.

2 Route A: quadratics over $\mathbb{Z}$ and the roots of $x^2 \equiv D$

Let $f = at^2 + bt + c$, $D = b^2 - 4ac$, $Q(h) = a^2h^2 - D$. For $p \nmid 2aD$ the roots of f modulo a split prime differ by $\pm\delta_p$, $\delta_p \equiv \sqrt{D}/a$. For d squarefree and coprime to $2aD$ with all prime factors split, the ordered pairs of roots modulo d correspond to sign patterns $\varepsilon \in \{-1, 0, 1\}^k$ (k the number of prime factors) with multiplicity $2^{k-j(\varepsilon)}$, j the number of non-zero entries, and the difference m_ε satisfies $m_\varepsilon \equiv \varepsilon_p \delta_p \pmod{p}$. Hence, with $d' = \prod_{\varepsilon_p \neq 0} p$, $m_\varepsilon \equiv 0 \pmod{d/d'}$ and $m_\varepsilon \pmod{d'}$ is a root of $x^2 \equiv D/a^2 \pmod{d'}$.

2.1 Reduction to root discrepancies

For $d \leq H$ write $R = d\{H/d\}$; then $\psi_d(m, H) = \mathbf{1}_{m \leq R} - \{H/d\}$ for $1 \leq m \leq d$, so

$$\sum_{s \neq s'} \psi_d(s' - s, H) = \Delta_d\left(\left\{\frac{H}{d}\right\}\right), \quad \Delta_d(\alpha) := \#\{\text{pairs} : m_{ss'} \leq \alpha d\} - \alpha \omega_f^\circ(d), \quad (2)$$

the discrepancy of the multiset of root differences at the point α , with $\omega_f^\circ(d) = \omega_f(d)^2 - \omega_f(d)$. By the correspondence above, for d coprime to $2aD$,

$$\Delta_d(\alpha) = \sum_{d'|d, d' > 1} 2^{k(d) - k(d')} \left(\#\{\nu \bmod d' : \nu^2 \equiv D/a^2, \overline{(d/d')} \nu \bmod d' \leq \alpha d'\} - \alpha \rho(d') \right),$$

where $\rho(d') = 2^{k(d')}$ is the number of roots and $\overline{(d/d')}$ the inverse of d/d' modulo d' (the map $\nu \mapsto \nu(d/d') \overline{(d/d')}$ identifies the classes). Thus the off-diagonal of a quadratic is a weighted sum, over moduli d' , of discrepancies of the roots of a fixed quadratic congruence, dilated by a unit and evaluated at the point $\{H/d\}$.

Proposition 1. *For an irreducible quadratic f ,*

$$\text{Off}_f(H) = \sum_{d \leq H} W_f(d) \Delta_d\left(\left\{\frac{H}{d}\right\}\right) + T_f(H), \quad T_f(H) = \sum_{d > H} W_f(d) \left(\#\{\text{pairs} : m_{ss'} \leq H\} - \frac{H \omega_f^\circ(d)}{d} \right),$$

both sums converging absolutely.

2.2 Small moduli: Hooley's equidistribution

The Erdős–Turán inequality bounds the sup-discrepancy $D_{d'}^*$ of the roots of $x^2 \equiv D/a^2$ modulo d' (dilated by a unit) by $\rho(d')/K + \sum_{k \leq K} k^{-1} |\sum_{\nu} e(k \bar{u} \nu / d')|$. Hooley [4] proved, as the key lemma of his asymptotic for $\sum_{n \leq x} \tau(n^2 + D)$, that the Weyl sums $\sum_{m \leq x} \sum_{\nu^2 \equiv D(m)} e(k \nu / m)$ have a power saving over the trivial bound $\sum_{m \leq x} \rho(m) \asymp x$, uniformly in k in a suitable range, by parametrisng the solutions (ν, m) of $\nu^2 - D = m\ell$ and applying Weil's bound for Kloosterman sums; the same argument gives the statement with the unit twist $\bar{u}$ and with weights that are multiplicative and bounded on average. Since $|W_f(d)| \asymp \lambda(d)/d$ with λ multiplicative and $\lambda(p) = p/(p-4)$ on split primes, partial summation of a bound $\sum_{d \leq x} \lambda(d) D_d^* \ll x(\log x)^{-1-\eta}$ gives

Proposition 2 (conditional). *If $\sum_{d \leq x} \lambda(d) D_d^* \ll x(\log x)^{-1-\eta}$ for some $\eta > 0$, then $\sum_{d \leq H} W_f(d) \Delta_d(\{H/d\}) = O_f(1)$.*

Hooley's lemma gives this bound with a power saving for the unweighted discrepancy of the roots of $x^2 \equiv D \pmod{m}$ over all $m \leq x$; the extension to the weights λ (bounded by a divisor-type function) and to the dilations by units is routine in his method, since the Kloosterman-sum bound is uniform in the residue class. We have not rewritten his proof in this generality and record it as the first of the two estimates needed.

2.3 Large moduli: the hyperbola

For $d > H$ only differences $m \leq H$ count. A pair with difference $m = h$ ($1 \leq h \leq H$) modulo d exists iff $\nu_f(d, h) \neq 0$, i.e. iff every prime factor of d divides $hQ(h)$; hence

$$T_f(H) = \sum_{h \leq H} \sum_{\substack{d > H \\ d | \text{rad}(hQ(h))}} W_f(d) \nu_f(d, h) - H \sum_{d > H} \frac{W_f(d) \omega_f^\circ(d)}{d}.$$

Both terms are of size $\asymp \log H$ (the second by the Dirichlet series $\sum_d W_f(d) \omega_f(d)^2 d^{-s} \sim c \zeta(s+1)^2$; the first because $\sum_{h \leq H} \tau(Q(h)) \asymp H \log H$), so their difference is a genuine secondary-term problem. Writing $W_f(d) = P(1) d^{-1} \prod_{p|d} p/(p-4)$ on split moduli and $d = Q(h)/e$ with the complementary divisor $e < Q(h)/H \leq a^2 H + |D|$, the first term becomes

$$P(1) \sum_{h \leq H} \frac{1}{Q(h)} \sum_{\substack{e|Q(h) \\ e < Q(h)/H}} e \lambda(Q(h)/e) \quad (+\text{the terms with } p \mid h),$$

a sum over the *small* divisors e of the quadratic polynomial $Q(h)$: this is the structure of Hooley's problem $\sum_{n \leq x} \tau(n^2 + D)$, where the divisors $d > x$ are handled through $e = (n^2 + D)/d < x$ and the count $\#\{h \leq H : e \mid Q(h)\} = H \rho(e)/e + O(\text{discrepancy})$ is again controlled by the equidistribution of the roots of Q modulo e . The only new feature is the weight $\lambda(Q(h)/e)$, which depends on the cofactor; writing $\lambda = \sum_{g|e} \kappa(g)$ with $\kappa(p) = 4/(p-4)$ reduces it to sums of the same type with e replaced by eg . We therefore expect, but have not carried out,

Proposition 3 (expected). $T_f(H) = O_f(1)$ for every irreducible quadratic f , by Hooley's method applied to the small divisors of $Q(h)$ with the multiplicative weight λ .

Propositions 2 and 3 together give Hypothesis (E) for quadratics, and hence (with Theorem 6 of the paper) the conjecture in Cesàro form. Numerically, $\text{Off}_f(H)$ is bounded for the 25 quadratics tested to $H = 10^5$ (paper, Table 2). The work remaining is the transcription of [4] with weights; it is not conceptual.

3 Route B: the function field $\mathbb{F}_q[u]$

Let $\mathcal{A} = \mathbb{F}_q[u]$, q odd, and $f \in \mathcal{A}[t]$ irreducible in t , separable, without fixed prime divisor. For a prime P of $\mathcal{A}$ write $|P| = q^{\deg P}$, $\omega_f(P)$ for the number of roots of f in $\mathcal{A}/P$, and define $\nu_f(d, h)$, $C(f)$, $S_f(h)$, b , P_P , W_f , a_f exactly as over $\mathbb{Z}$ with $|P|$ in place of p ; all products converge absolutely as before. " $h \leq H$ " becomes " $h \neq 0, \deg h < N$ ", a set of $q^N - 1$ elements, and $\log H$ becomes N .

Theorem 4. With $a_f(d) = W_f(d) \omega_f(d)$ and $P(1) = \prod_P P_P = W_f(1)$, for every $N \geq 1$,

$$\sum_{\substack{h \neq 0 \\ \deg h < N}} \left(\frac{S_f(h)}{C^2} - 1 \right) = - \sum_{1 \leq \deg d \leq N} a_f(d) - q^N \sum_{\deg d > N} \frac{a_f(d)}{|d|} + (1 - P(1)) + \text{Off}_f(N),$$

$$\text{Off}_f(N) = \sum_{\deg d > N} W_f(d) \sum_{s \neq s'} \left(\mathbf{1}_{0 \leq \deg \tilde{m} < N} - q^{N - \deg d} \right),$$

$\tilde{m}$ the representative of $s' - s$ of degree $< \deg d$. In particular the pairs of distinct roots modulo d contribute nothing for $\deg d \leq N$: Hypothesis (E) is empty for small moduli.

Proof. The expansion $S_f(h)/C^2 = \sum_Q b(Q) c_Q^f(h)$ over squarefree monic Q holds verbatim, and for $Q \neq 1$, $\sum_{h \neq 0, \deg h < N} c_Q^f(h) = \sum_{d|Q} |d| \mu(Q/d) \omega_f(Q/d)^2 \sum_{s, s' \bmod d} \psi_d(s' - s)$ with $\psi_d(m) = \#\{h \neq 0, \deg h < N : h \equiv m \pmod{d}\} - q^{N - \deg d}$ (the subtraction is allowed because $\sum_{d|Q} \mu(Q/d) = 0$). For $\deg d \leq N$ every class modulo d contains exactly $q^{N - \deg d}$ polynomials of degree $< N$, of which one is 0 iff $d \mid m$, so $\psi_d(m) = -\mathbf{1}_{d|m}$, and this includes $d = 1$: $\psi_1 = -1$, since $h = 0$ is excluded. For $\deg d > N$ a class contains at most one nonzero h of degree $< N$, so $\psi_d(m) = \mathbf{1}_{0 \leq \deg \tilde{m} < N} - q^{N - \deg d}$. Regrouping by d gives $\sum_d W_f(d) \sum_{s, s'} \psi_d(s' - s) + 1$, the $+1$ compensating the excluded term $Q = 1$, whose value in the regrouped sum is $\psi_1 = -1$. For $\deg d \leq N$ the inner sum is $-\omega_f(d)$ (the pairs with $s \equiv s'$), giving $-P(1) - \sum_{1 \leq \deg d \leq N} a_f(d)$; for $\deg d > N$ the diagonal pairs give $-\omega_f(d) q^{N - \deg d}$ and the off-diagonal pairs give $\text{Off}_f(N)$. (Over $\mathbb{Z}$ the modulus $d = 1$ drops out because $\#\{h \leq H\} = H$ exactly; here it leaves the constant $1 - P(1)$.) $\square$

Proposition 5. $P_N(d) = P(d) + O(q^{-(N-\deg d)/2} \cdot \text{polylog})$ and, with $D_f(s) = \sum_d a_f(d)|d|^{-s}$,

$$\sum_{1 \leq \deg d \leq N} a_f(d) = \frac{N}{C(f)} + c_f + O(q^{-N/2}),$$

because, with $T = q^{-s}$, $D_f(s) = \zeta_{\mathcal{A}}(s+1)L(s+1, \chi_D)E_f(s)$ where $\zeta_{\mathcal{A}}(s+1) = 1/(1-T)$, $L(s+1, \chi_D)$ is a polynomial in T , and $E_f(s) = \tilde{H}_f(s)(1-q^{-1}T^2)^2/L(2s+2, \chi_D)$ with $\tilde{H}_f$ absolutely convergent for $|T| < q$ and the zeros of the polynomial $L(2s+2, \chi_D)$ on $|T| = q^{3/4}$ by Weil; so D_f is meromorphic in $|T| < q^{3/4}$ with the single simple pole $T = 1$, at which $(1-T)D_f \rightarrow 1/C(f)$ by the same local computation as over $\mathbb{Z}$. (The error term is in fact $O(q^{-(3/4-\varepsilon)N})$.)

Thus, over $\mathbb{F}_q[u]$, the function-field form of the conjecture,

$$\sum_{\substack{h \neq 0 \\ \deg h < N}} (S_f(h) - C(f)^2) = -C(f)N + O_f(1), \quad (3)$$

is equivalent to $\text{Off}_f(N) = O(1)$: the whole content of Hypothesis (E) sits in the off-diagonal pairs modulo the moduli of degree $> N$ (the diagonal tail $q^N \sum_{\deg d > N} a_f(d)/|d|$ tends to $1/(C(f)(q-1))$). The coefficient is $-C$ rather than $-\frac{1}{2}C$ because the fractional parts $\{H/d\}$, whose mean $\frac{1}{2}$ produces the $\frac{1}{2}$ over $\mathbb{Z}$, are replaced by the exact value 1: a modulus of degree $\leq N$ misses exactly the class of 0.

3.1 The remainder as a finite degree-distribution problem

For $\deg d > N$ a residue class m modulo d contains at most one nonzero h of degree $< N$, namely its representative $\tilde{m}$ when $0 \leq \deg \tilde{m} < N$. For a quadratic $f = t^2 - D$ this has a strong consequence: if (s, s') , $s \neq s'$, has $0 \leq \deg \tilde{m} < N$, split $d = d_1 d_2$ with $d_2 = \prod \{P \mid d : s \equiv s' \pmod{P}\}$; then $d_2 \mid \tilde{m}$, so $\deg d_2 \leq N-1$, and for $P \mid d_1$ the roots are $\pm\sqrt{D}$, so $d_1 \mid \tilde{m}^2 - 4D \neq 0$ and $\deg d_1 \leq \max(2N-2, \deg D)$. Hence

$$\text{Off}_f(N) = \sum_{N < \deg d \leq M(N)} W_f(d) A_N(d) - q^N \sum_{\deg d > N} \frac{W_f(d)(\omega_f(d)^2 - \omega_f(d))}{|d|}, \quad M(N) = N-1 + \max(2N-2, \deg D)$$

with $A_N(d)$ the number of ordered pairs of distinct roots whose difference has degree $< N$; the series is computable from $\sum_d W_f(d)\omega_f(d)^2/|d| = 1$ and $\sum_d W_f(d)\omega_f(d)/|d| = \prod_P (1 + b_P(\omega_P - \omega_P^2))$ (both by $\sum_{d|Q} \mu(Q/d) = 0$). So the whole question is whether, over the moduli of degree between N and about $3N$, the differences $\pm 2\sqrt{D}$ of the roots modulo d fall among the residues of degree $< N$ with frequency $q^{N-\deg d}$ up to a bounded total discrepancy. For a prime modulus P of degree k , writing the indicator of $\deg \tilde{x} < N$ through the additive characters trivial on $\mathcal{A}_{<N}$,

$$\mathbf{1}_{\deg \tilde{x} < N} = q^{N-k} \sum_{\deg a < k-N} \psi\left(\frac{ax}{P}\right),$$

the count becomes a sum over $a \neq 0$ of the Weyl sums $\sum_{\deg P=k} \psi(2a\sqrt{D}/P)$ of the roots of a quadratic congruence over the primes of degree k : the function-field analogue of the sums of Duke, Friedlander and Iwaniec [2]. Over $\mathbb{F}_q[u]$ such sums are governed by the monodromy of the relevant Kloosterman-type sheaf and Deligne's equidistribution theorem [5]: for fixed N, k and $q \rightarrow \infty$ each has square-root cancellation, $O(q^{k/2})$ against the trivial q^k/k , and since only the finitely many degrees $k \leq M(N)$ occur, the remainder is $O(q^{-1/2})$ in that regime.

Proposition 6 (expected, $q \rightarrow \infty$). *For a fixed irreducible quadratic $f = t^2 - D(u)$ and fixed N , $\text{Off}_f(N) = O_{N, \deg D}(q^{-1/2})$; hence (3) holds with the $O(1)$ replaced by $O(q^{-1/2})$ as $q \rightarrow \infty$.*

This is the function-field theorem to write out: its only input beyond Theorem 4 is the equidistribution of the roots of a quadratic congruence over primes of a given degree, which in the large- q regime is a standard consequence of Deligne’s theorem, in the same way as the results of Bary-Soroker and Entin [1, 3] on prime values of polynomials over $\mathbb{F}_q[u]$.

3.2 Exact computation

Theorem 4 and the finite-support formula can be checked exactly, and in two independent ways: from the $S_f(h)$ themselves (each $S_f(h)/(C^2P(1))$ is a finite product over the primes dividing $h(h^2-4D)$; `thesis/ff.ts`) and from the roots of f modulo every squarefree d of degree $\leq M(N)$ (`thesis/ff-tail.ts`, with a brute-force recomputation of the finite sum for the smallest cases). Table 1 reports the result for $q = 3, 5, 7$ and six choices of D (the cases D and $D(u+c)$ coincide, since $u \mapsto u+c$ is a degree-preserving automorphism, so only one representative of each translation class is listed). The two computations of $\text{Off}_f(N)$ agree to $3 \cdot 10^{-5}$; the increments of $\sum_{\deg d \leq N} a_f(d)$ converge to $1/C(f)$ as Proposition 5 predicts, after a transient for $q = 3$ caused by the primes of norm 3 with $\omega_f(P) = 2$, where $P_P = 1 - 4/(3-2)^2 = -3$ and $P(1)$ is negative; the diagonal tail converges to $1/(C(f)(q-1))$; and $\text{Off}_f(N)$ is bounded, $|\text{Off}_f(N)| \leq 1.83$ with a variation over N of at most 1.13, with no sign of growth, which is what (3) requires.

4 Assessment

Route B is the shorter path to a theorem: Theorem 4 is proved, Proposition 5 is a standard consequence of Weil’s theorem, and Proposition 6 needs one equidistribution input that exists in the literature in the large- q regime. Route A is the harder and more valuable path, since it concerns the integers; it is a transcription of Hooley’s 1963 method with multiplicative weights, and we have identified exactly where the two estimates (Propositions 2 and 3) enter. Neither is done here.

References

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Table 1: Theorem 4 for $f = t^2 - D$ over $\mathbb{F}_q[u]$: LHS, $\sum_{1 \leq \deg d \leq N} a_f(d)$ and its increment (limit $1/C(f)$), the diagonal tail $q^N \sum_{\deg d > N} a_f(d)/|d|$ (limit $1/(C(f)(q-1))$), and $\text{Off}_f(N)$.

q	D	$C(f)$	$1 - P(1)$	N	$\#h$	LHS	$\sum a_f$	incr.	diag. tail	$\text{Off}_f(N)$
3	u	0.851	2.999	1	2	-2.000	1.999	—	1.168	-1.832
				2	8	-3.202	5.198	3.199	0.305	-0.699
				3	26	-4.758	5.302	0.104	0.810	-1.645
				4	80	-5.855	7.181	1.879	0.551	-1.122
				5	242	-7.167	8.308	1.126	0.527	-1.331
				6	728	-8.201	9.252	0.945	0.637	-1.311
				7	2186	-9.548	10.587	1.334	0.576	-1.385
3	$u^2 + 1$	1.459	3.857	1	2	-2.000	5.714	—	-0.094	-0.237
				2	8	-2.286	5.306	-0.408	0.127	-0.710
				3	26	-2.921	5.129	-0.177	0.558	-1.092
				4	80	-3.530	6.522	1.394	0.279	-0.586
				5	242	-4.214	6.994	0.472	0.366	-0.711
				6	728	-4.925	7.770	0.776	0.322	-0.691
				7	2186	-5.626	8.384	0.615	0.351	-0.748
3	$u^2 + 2$	1.703	0.500	1	2	-0.801	1.000	—	0.332	0.030
				2	8	-1.803	1.699	0.700	0.295	-0.309
				3	26	-2.019	2.273	0.574	0.311	0.065
				4	80	-2.827	2.924	0.651	0.282	-0.121
				5	242	-3.372	3.476	0.551	0.294	-0.103
				6	728	-3.949	4.061	0.585	0.297	-0.091
				7	2186	-4.528	4.661	0.600	0.292	-0.075
5	u	0.956	0.721	1	4	-1.768	1.209	—	0.366	-0.914
				2	24	-2.958	2.803	1.594	0.237	-0.638
				3	124	-3.770	3.728	0.925	0.262	-0.501
				4	624	-4.733	4.774	1.046	0.266	-0.414
				5	3124	-5.769	5.845	1.071	0.258	-0.388
5	$u^2 + 2$	1.247	0.701	1	4	-0.978	1.197	—	0.287	-0.194
				2	24	-2.250	2.493	1.296	0.142	-0.316
				3	124	-2.786	2.986	0.493	0.215	-0.286
				4	624	-3.701	3.857	0.871	0.202	-0.342
				5	3124	-4.465	4.671	0.813	0.197	-0.298
7	u	0.979	0.434	1	6	-1.474	1.245	—	0.196	-0.468
				2	48	-2.392	2.451	1.207	0.167	-0.208
				3	342	-3.478	3.455	1.003	0.169	-0.288
				4	2400	-4.439	4.468	1.013	0.171	-0.235

Table 2: The finite-support formula checked: finite sum, series, their difference, and $\text{Off}_f(N)$ from Table 1.

q	D	N	$M(N)$	$\#d$	finite sum	series	difference	$\text{Off}_f(N)$ from LHS
3	u	1	1	0	0.0000	1.8322	-1.8322	-1.8322
		2	3	6	1.5993	2.2980	-0.6988	-0.6988
		3	6	199	1.9465	3.5913	-1.6448	-1.6448
		4	9	4726	1.7928	2.9146	-1.1218	-1.1218
		5	12	112209	2.3993	3.7307	-1.3314	-1.3314
3	$u^2 + 1$	1	2	1	0.0000	0.2367	-0.2367	-0.2367
		2	3	5	0.0000	0.7102	-0.7102	-0.7102
		3	6	146	1.2164	2.3082	-1.0918	-1.0918
		4	9	3500	0.9700	1.5557	-0.5857	-0.5857
		5	12	83301	1.1099	1.8208	-0.7110	-0.7110
3	$u^2 + 2$	1	2	2	0.1999	0.1696	0.0303	0.0303
		2	3	6	0.0000	0.3088	-0.3088	-0.3088
		3	6	199	0.4179	0.3528	0.0650	0.0650
		4	9	4726	0.2862	0.4071	-0.1209	-0.1209
		5	12	112209	0.4283	0.5308	-0.1025	-0.1025
5	u	1	1	0	0.0000	0.9141	-0.9141	-0.9141
		2	3	33	0.1063	0.7447	-0.6384	-0.6384
		3	6	4032	0.7041	1.2048	-0.5006	-0.5007
5	$u^2 + 2$	1	2	5	0.6271	0.8215	-0.1943	-0.1943
		2	3	28	0.1140	0.4301	-0.3160	-0.3160
		3	6	3314	0.6868	0.9730	-0.2862	-0.2862
7	u	1	1	0	0.0000	0.4679	-0.4679	-0.4679
		2	3	96	0.3520	0.5597	-0.2077	-0.2077
		3	6	29328	0.4140	0.7022	-0.2882	-0.2882